

机器学习导论

第11章 主成分分析与非负矩阵分解



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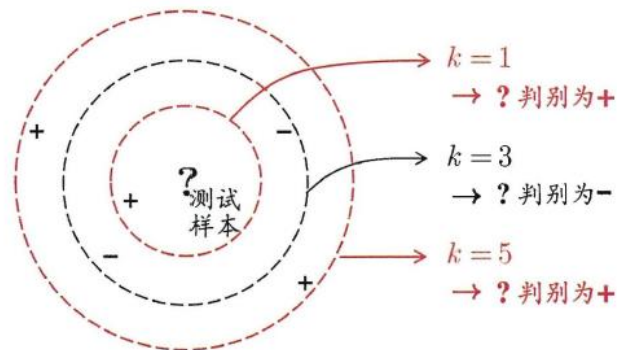
03.非负矩阵分解(Non-negative Matrix Factorization)

1. 距离、空间和分类器

- 从可视化角度，特征可以看做是空间的维度，特征的定义其实也是空间的定义。定义数据的各个维度之间很难保证相互独立、不相关、不冗余
- 很多分类器是基于距离（或相似度）的
 - 距离的计算是依赖特征（或空间定义）的
- 最典型的基于距离的分类器是k最近邻分类器

1. kNN分类器

- 最直观的分类器
- 无(显式)训练
 - 训练数据即分类器
 - 距离（或相似度）是关键
- 参数：k、空间、度量
- 泛化错误率不超过贝叶斯最优分类器的错误率的2倍



1. kNN分类器

- 算法
 - 根据距离度量，在训练数据集中找到与 x 最邻近的 k 个样本；
 - 根据分类规则（如多数投票）判定 x 的类别 y

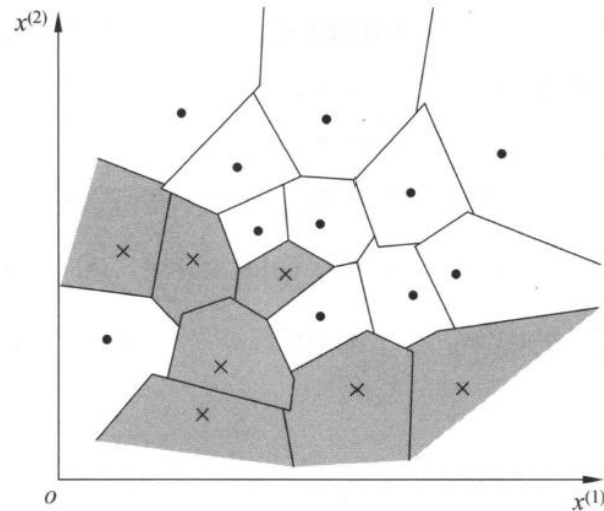


图 3.1 k 近邻法的模型对应特征空间的一个划分

1. 距离度量

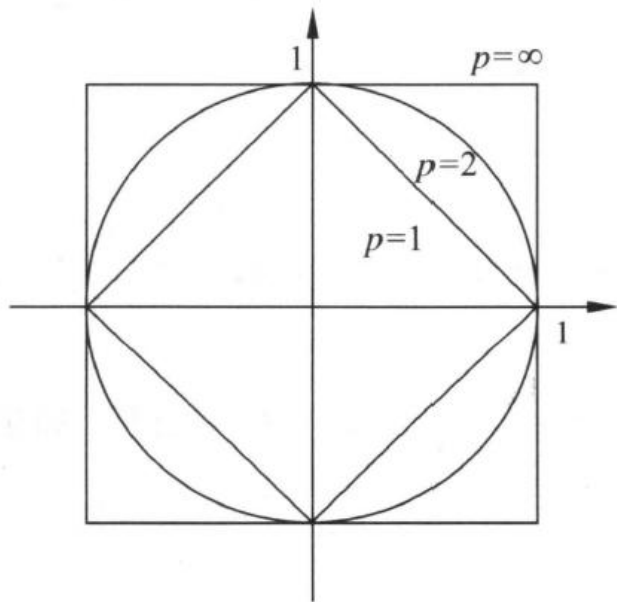


图 3.2 L_p 距离间的关系

$$L_p(x_i, x_j) = \left(\sum_{l=1}^n |x_i^{(l)} - x_j^{(l)}|^p \right)^{\frac{1}{p}}$$

$$L_2(x_i, x_j) = \left(\sum_{l=1}^n |x_i^{(l)} - x_j^{(l)}|^2 \right)^{\frac{1}{2}}$$

$$L_1(x_i, x_j) = \sum_{l=1}^n |x_i^{(l)} - x_j^{(l)}|$$

与原点的 L_p 距离为1的点组成的图形。

1. 相似度

- Cosine Similarity
- Jaccard Similarity (集合相似度)

表 6.1 常用核函数

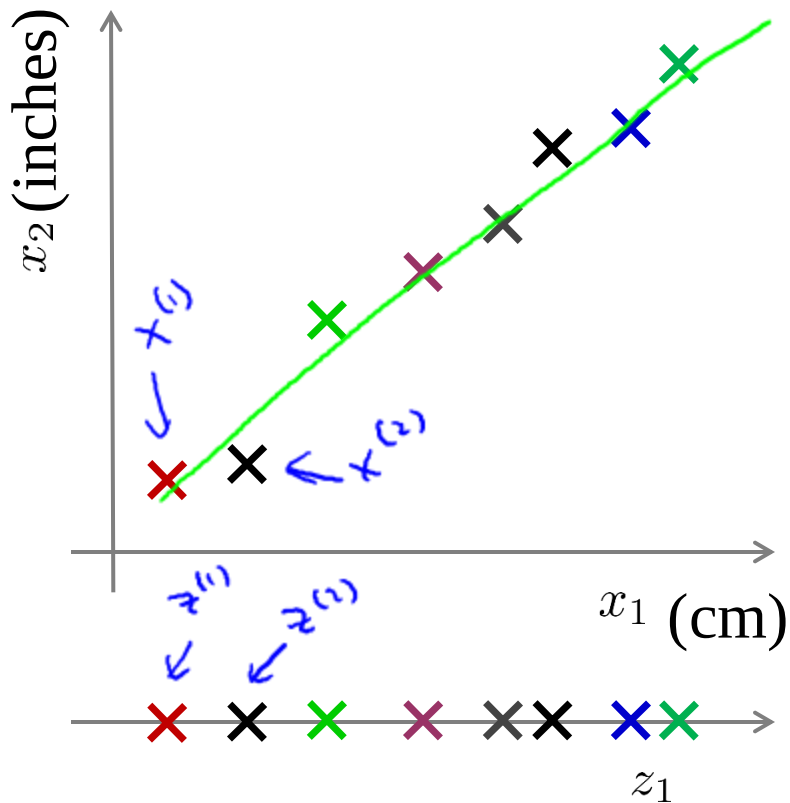
名称	表达式	参数
线性核	$\kappa(\mathbf{x}_i, \mathbf{x}_j) = \mathbf{x}_i^T \mathbf{x}_j$	
多项式核	$\kappa(\mathbf{x}_i, \mathbf{x}_j) = (\mathbf{x}_i^T \mathbf{x}_j)^d$	$d \geq 1$ 为多项式的次数
高斯核	$\kappa(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(-\frac{\ \mathbf{x}_i - \mathbf{x}_j\ ^2}{2\sigma^2}\right)$	$\sigma > 0$ 为高斯核的带宽(width)
拉普拉斯核	$\kappa(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(-\frac{\ \mathbf{x}_i - \mathbf{x}_j\ }{\sigma}\right)$	$\sigma > 0$
Sigmoid 核	$\kappa(\mathbf{x}_i, \mathbf{x}_j) = \tanh(\beta \mathbf{x}_i^T \mathbf{x}_j + \theta)$	$\tanh$ 为双曲正切函数, $\beta > 0, \theta < 0$

$d = 1$ 时退化为线性核.
高斯核亦称 RBF 核.

2. 主成分分析 (PCA, Principle Component Analysis)

- 特征表示
 - 特征空间重构
 - 面向任务：维持数据间关系，并方差最大化
- 主要用途
 - 数据压缩 (Data Compression)
 - 数据可视化 (Data Visualization)
 - 维度压缩 (Dimension Reduction)

2.1.1 数据压缩



Reduce data from
2D to 1D

$$x^{(1)} \in \mathbb{R}^2 \rightarrow z^{(1)} \in \mathbb{R}$$

$$x^{(2)} \in \mathbb{R}^2 \rightarrow z^{(2)} \in \mathbb{R}$$

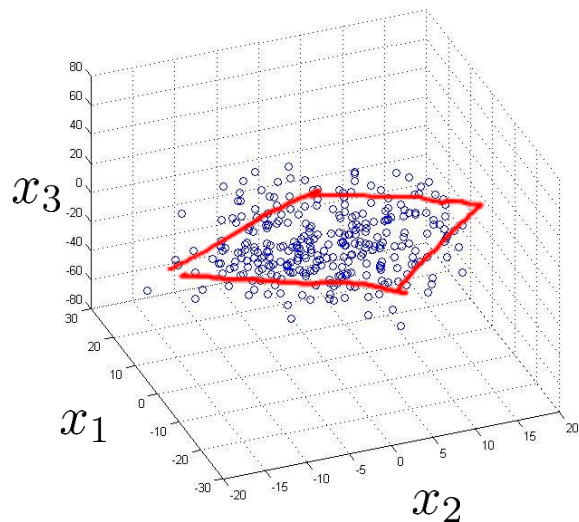
⋮

$$x^{(m)} \in \mathbb{R}^2 \rightarrow z^{(m)} \in \mathbb{R}$$

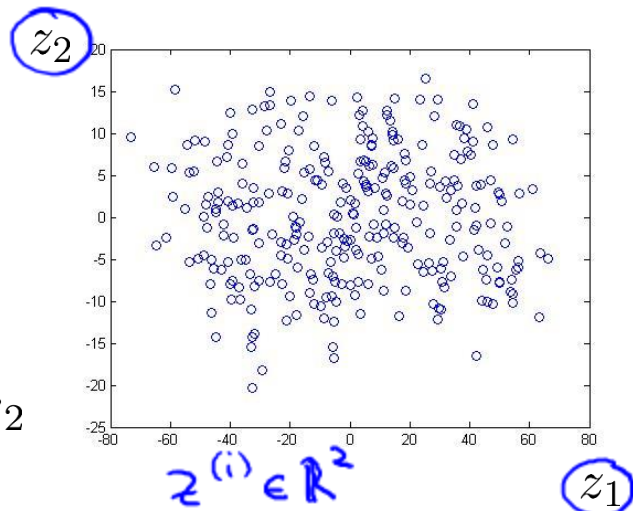
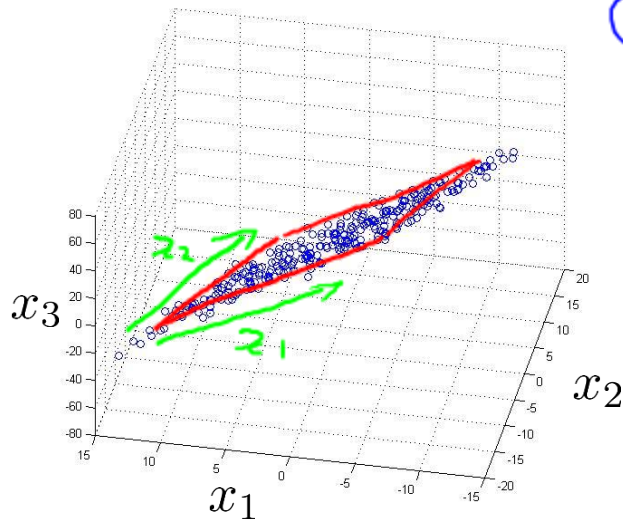
2.1.1 数据压缩

10000 → 1000

Reduce data from 3D to 2D



$$x^{(i)} \in \mathbb{R}^3$$



$$z^{(i)} \in \mathbb{R}^2$$

$$z = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} \quad z^{(i)} = \begin{bmatrix} z_1^{(i)} \\ z_2^{(i)} \end{bmatrix}$$

2013年9月, 2日欧洲信用卡欺诈检测, 28列特征 (492/28万)

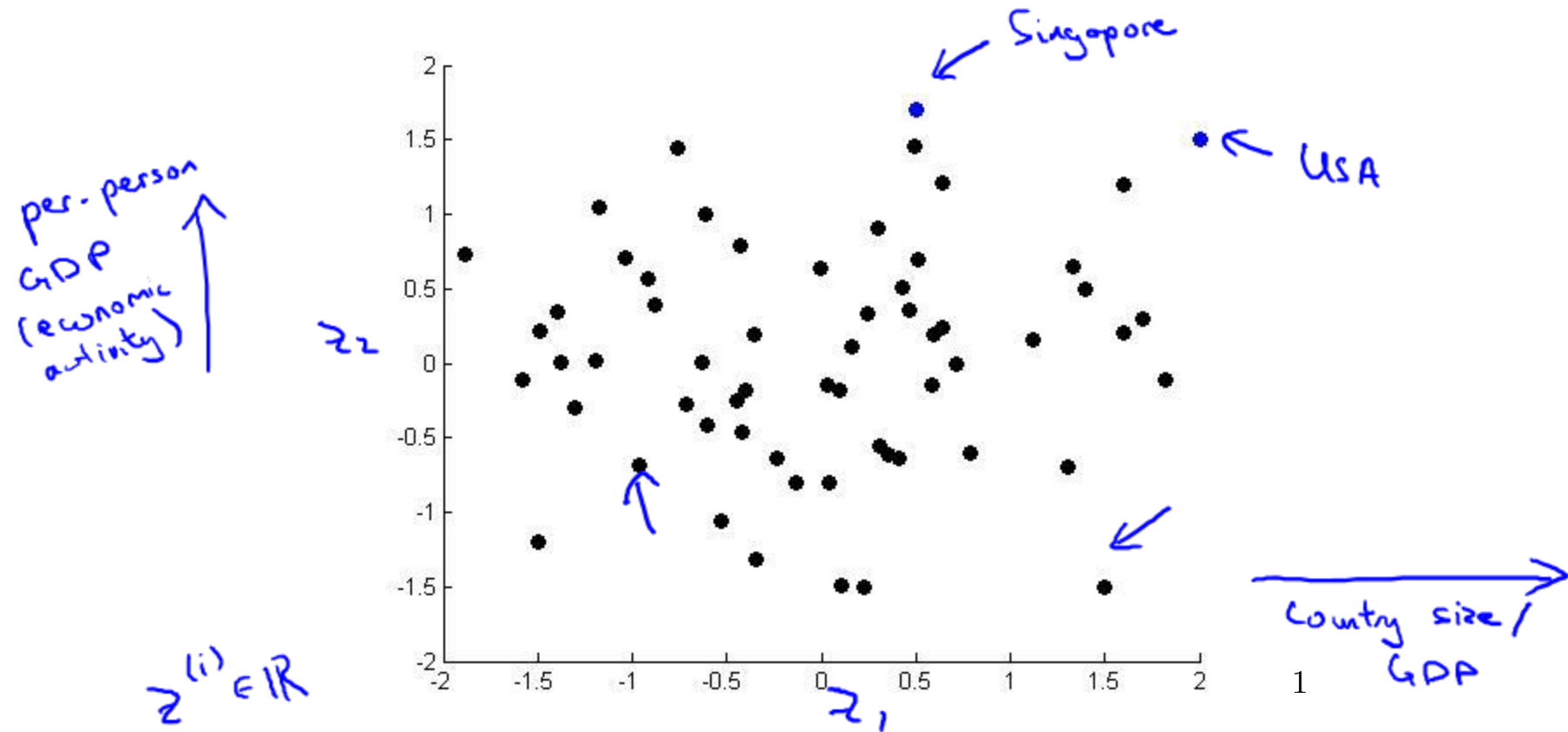
2.1.2 数据可视化

Country	GDP (trillions of US\$)	Per capita GDP (thousands of intl. \$)	Human Development Index	Life expectancy	Poverty Index (Gini as percentage)	Mean household income (thousands of US\$)	...
Canada	1.577	39.17	0.908	80.7	32.6	67.293	...
China	5.878	7.54	0.687	73	46.9	10.22	...
India	1.632	3.41	0.547	64.7	36.8	0.735	...
Russia	1.48	19.84	0.755	65.5	39.9	0.72	...
Singapore	0.223	56.69	0.866	80	42.5	67.1	...
USA	14.527	46.86	0.91	78.3	40.8	84.3	...
...	...	...	...	...	...	...	...

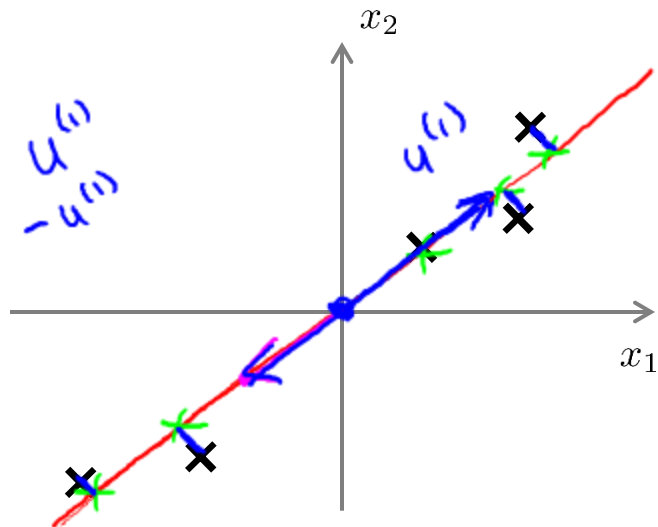
2.1.2 数据可视化

Country	z_1	z_2	$z^{(i)} \in \mathbb{R}^2$
Canada	1.6	1.2	
China	1.7	0.3	Reduce data from 500 to 2D
India	1.6	0.2	
Russia	1.4	0.5	
Singapore	0.5	1.7	
USA	2	1.5	
...	...	...	

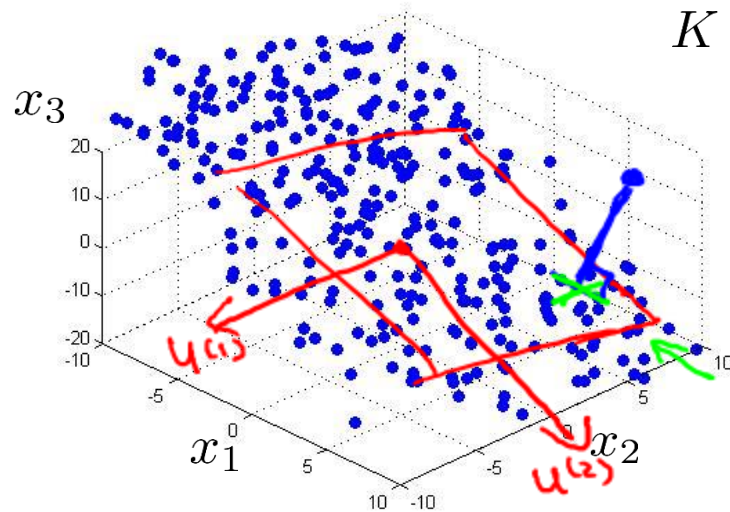
2.1.2 在PCA获得两个主成分后将各国家的数据可视化



2.1.3 PCA进行特征维度压缩的示例



$3D \rightarrow 2D$
 $K = 2$



Reduce from 2-dimension to 1-dimension: Find a direction (a vector $\underline{u^{(1)}} \in \mathbb{R}^n$) onto which to project the data so as to minimize the projection error.

Reduce from n-dimension to k-dimension: Find k vectors $\underline{u^{(1)}, u^{(2)}, \dots, u^{(k)}}$ onto which to project the data, so as to minimize the projection error.

2.2 主成分分析的主要目标和思路

- 删掉重复特征
- 去除冗余特征
- 建立尽可能少的特征，使得这些新特征两两不相关
- 通过线性组合现有特征实现组合后的方差(或离散程度)最大化

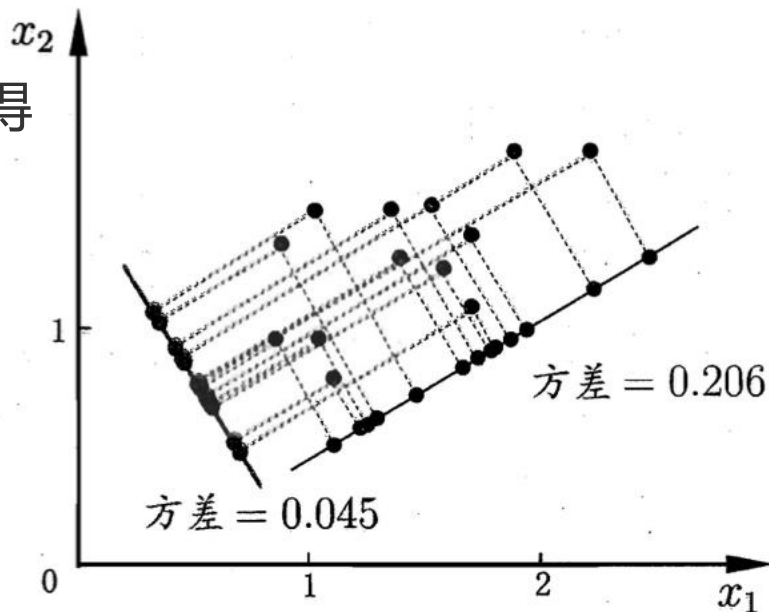


图 10.4 使所有样本的投影尽可能分开(如图中红线所示), 则需最大化投影点的方差

2.2 PCA算法-1. 数据预处理(Preprocessing)

输入数据: $x^{(1)}, x^{(2)}, \dots, x^{(m)}$

预处理 (对各维进行平移和拉伸 feature scaling/mean normalization):

1. 计算第j维的均值 $\mu_j = \frac{1}{m} \sum_{i=1}^m x_j^{(i)}$

2. 平移样本第j维至原点 $x_j - \mu_j$

3. 拉伸第j维至同等规模, 如 $\frac{x_j - \mu_j}{\sigma_j}$

2.2 PCA算法-2. 主算法

计算各维度间的协方差矩阵

$$\Sigma = \frac{1}{m} \sum_{i=1}^n (x^{(i)})(x^{(i)})^T \quad \text{计算维度间的协方差矩阵}$$

计算维度协方差矩阵 Σ 的特征向量(eigenvector)

```
[U,S,V] = svd(Sigma); // 常用  
[V,D]=eig(Sigma);    // 也可用
```

求解协方差矩阵的特征值、特征向量就是在相互正交的维度上最大化方差。

特征向量：表征变换后的新的方向

2.3 PCA的优化目标

$\mathbf{x}_i$ 在新空间中超平面上的投影是 $\mathbf{W}^T \mathbf{x}_i$, 若所有样本点的投影能尽可能分开, 则应该使投影后样本点的方差最大化, 如图 10.4 所示.

投影后样本点的方差是 $\sum_i \mathbf{W}^T \mathbf{x}_i \mathbf{x}_i^T \mathbf{W}$, 于是优化目标可写为

$$\begin{aligned} \max_{\mathbf{W}} \quad & \text{tr}(\mathbf{W}^T \mathbf{X} \mathbf{X}^T \mathbf{W}) \\ \text{s.t.} \quad & \mathbf{W}^T \mathbf{W} = \mathbf{I}, \end{aligned} \quad (10.16)$$

- 优化目标: 最大化投影后样本点的方差
- 约束: 新方向相互正交

2.3 最大化投影后方差与特征值分解

对式(10.15)或(10.16)使用拉格朗日乘子法可得

$$\mathbf{X}\mathbf{X}^T\mathbf{W} = \lambda\mathbf{W}, \quad (10.17)$$

于是, 只需对协方差矩阵 $\mathbf{X}\mathbf{X}^T$ 进行特征值分解, 将求得特征值排序: $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d$, 再取前 d' 个特征值对应的特征向量构成 $\mathbf{W} = (\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_{d'})$. 这就是主成分分析的解. PCA 算法描述如图 10.5 所示.

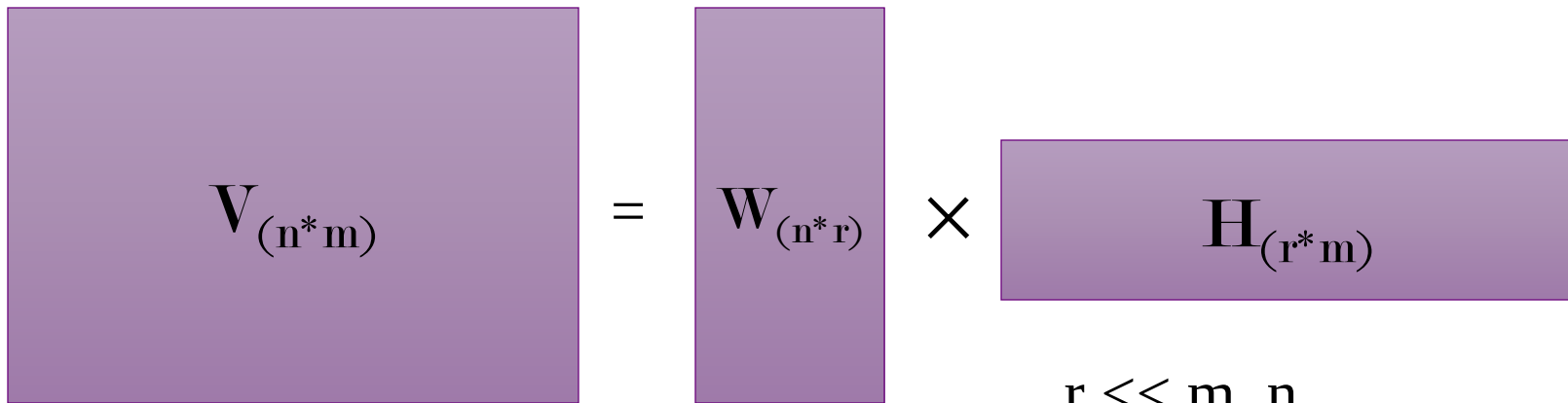
3. 非负矩阵分解(Non-negative Matrix Factorization)

- Learning the parts of objects by nonnegative matrix factorization (Nature, 1999)
- D.D. Lee (Bell Lab. 现在MIT脑认知科学实验室负责人), H.S. Seung (from MIT)
- 功能：特征压缩、软聚类
- 应用领域：图像特征抽取、文本语义模型、基因数据分析、语音信号处理、商品推荐等

3.非负矩阵分解

- 给定非负矩阵V, 将其分解为非负矩阵W和H如下:

$$V_{(n \times m)} \approx W_{(n \times r)} H_{(r \times m)}$$



$$V_{(n \times m)} = W_{(n \times r)} \times H_{(r \times m)}$$

$$r \ll m, n$$

$$(m+n) * r < nm$$

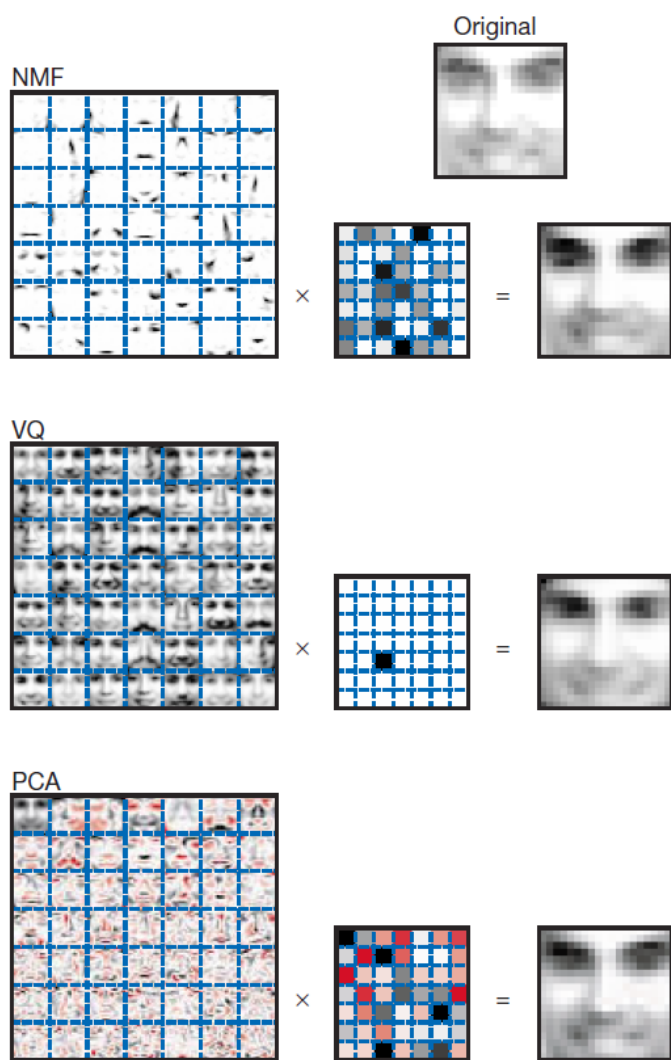
在图像压缩上的应用

Figure 1 Non-negative matrix factorization (NMF) learns a parts-based representation of faces, whereas vector quantization (VQ) and principal components analysis (PCA) learn holistic representations. The three learning methods were applied to a database of $m = 2,429$ facial images, each consisting of $n = 19 \times 19$ pixels, and constituting an $n \times m$ matrix V . All three find approximate factorizations of the form $V \approx WH$, but with three different types of constraints on W and H , as described more fully in the main text and methods. As shown in the 7×7 montages, each method has learned a set of $r = 49$ basis images. Positive values are illustrated with black pixels and negative values with red pixels. A particular instance of a face, shown at top right, is approximately represented by a linear superposition of basis images. The coefficients of the linear superposition are shown next to each montage, in a 7×7 grid, and the resulting superpositions are shown on the other side of the equality sign. Unlike VQ and PCA, NMF learns to represent faces with a set of basis images resembling parts of faces.

V : $n \times m$, n 行: 像素, m 列: 一幅图像

W : $n \times r$, r 列, 每列一个basis image

H : $r \times m$, m 列, 每列一个encoding



在文本上的应用



- 每个basis topic为W中该列出现次数最多的8个单词，颜色表示程度
- 一词多义被成功分在不同basis topic中
- Lead 领导; 铅

Figure 4 Non-negative matrix factorization (NMF) discovers semantic features of $m = 30,991$ articles from the Grolier encyclopedia. For each word in a vocabulary of size $n = 15,276$, the number of occurrences was counted in each article and used to form the $15,276 \times 30,991$ matrix V . Each column of V contained the word counts for a particular article, whereas each row of V contained the counts of a particular word in different articles. The matrix was approximately factorized into the form WH using the algorithm described in Fig. 2. Upper left, four of the $r = 200$ semantic features (columns of W). As they are very high-dimensional vectors, each semantic feature is represented by a list of the eight words with highest frequency in that feature. The darkness of the text indicates the relative frequency of each word within a feature. Right, the eight most frequent words and their counts in the encyclopedia entry on the 'Constitution of the United States'. This word count vector was approximated by a superposition that gave high weight to the upper two semantic features, and none to the lower two, as shown by the four shaded squares in the middle indicating the activities of H . The bottom of the figure exhibits the two semantic features containing 'lead' with high frequencies. Judging from the other words in the features, two different meanings of 'lead' are differentiated by NMF.

Encyclopedia entry:
'Constitution of the
United States'

president (148)
congress (124)
power (120)
united (104)
constitution (81)
amendment (71)
government (57)
law (49)

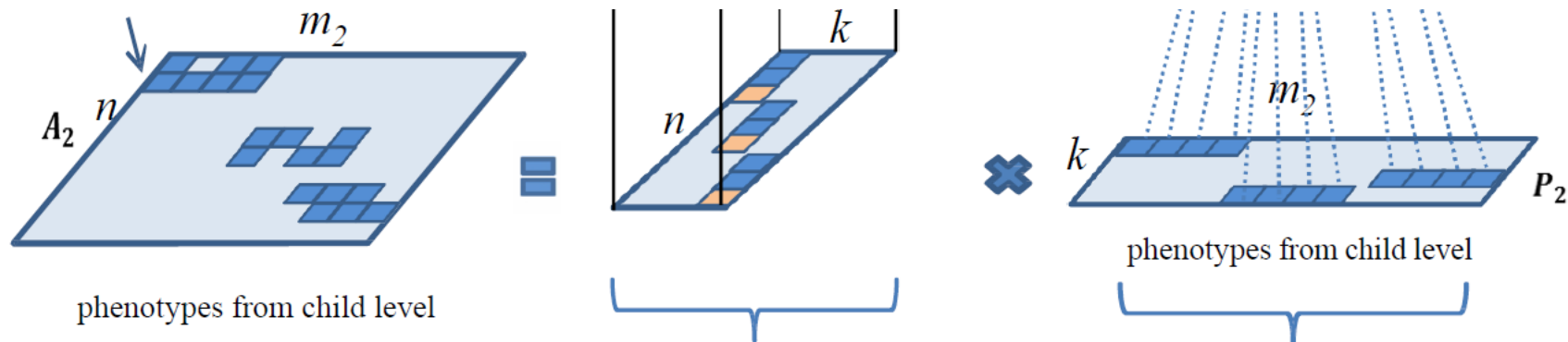


court government council culture supreme constitutional rights justice	president served governor secretary senate congress presidential elected
flowers leaves plant perennial flower plants growing annual	disease behaviour glands contact symptoms skin pain infection

metal process method paper ... glass copper lead steel

person example time people ... rules lead leads law

在致病基因(疾病-基因关联)预测上的应用



- V : 表型-基因矩阵, 用户-商品矩阵
- 分解后
 - W : 表型-簇, 用户-簇
 - H : 压缩特征-基因, 压缩特征-商品

• 分解前后的误差: $V = WH + E$

• 损失函数: $\min_{W,H} \|V - WH\|$

使用欧式距离衡量误差 $\min_{W,H} \sum_{i,j} (V_{ij} - (WH)_{ij})^2$

包含KL散度的误差 $\min_{W,H} \sum_{i,j} (V_{ij} \log \frac{V_{ij}}{(WH)_{ij}} - V_{ij} + (WH)_{ij})$

3.1 优化：最小化分解前后误差

- 假设噪声E服从高斯分布：

$$p(E|W, H) = \frac{1}{\sqrt{2\pi}\sigma_{ij}} \exp\left(-\frac{E_{ij}^2}{2\sigma_{ij}^2}\right)$$

- 构造似然函数：

$$\{W, H\} = \operatorname{argmax}_{W, H} p(V|W, H) = \operatorname{argmin}_{W, H} \{-\log p(V|W, H)\}$$

$$L(W, H) = \frac{1}{2\sigma_{ij}^2} \sum_{i,j} (V_{ij} - (WH)_{ij})^2 + \sum_{i,j} \log(\sqrt{2\pi}\sigma_{ij})$$

$$L(W, H) = \frac{1}{2\sigma_{ij}^2} \sum_{i,j} (V_{ij} - (WH)_{ij})^2 + \sum_{i,j} \log(\sqrt{2\pi}\sigma_{ij})$$

$$\begin{aligned} \frac{\partial L(W, H)}{\partial W_{ik}} &= c \left[\sum_j H_{kj} (V_{ij} - (WH)_{ij}) \right] \\ &= c \left[\sum_j V_{ij} H_{kj} - \sum_j (WH)_{ij} H_{kj} \right] \\ &= c \left[(VH^T)_{ik} - (WHH^T)_{ik} \right] \end{aligned}$$

$$\begin{aligned} \frac{\partial L(W, H)}{\partial H_{kj}} &= c \left[\sum_i W_{ik} V_{ij} - \sum_i (WH)_{ij} W_{ik} \right] \\ &= c \left[(W^T V)_{kj} - (W^T W H)_{kj} \right] \end{aligned}$$

根据优化结果为梯度下降提供学习步长

从负梯度方向进行更新

$$W_{ik}^{(n+1)} \leftarrow W_{ik}^{(n)} + \lambda_1 * [(VH^T)_{ik} - (WHH^T)_{ik}]$$

$$H_{kj}^{(n+1)} \leftarrow H_{kj}^{(n)} + \lambda_2 * [(W^TV)_{kj} - (W^TW H)_{kj}]$$

设学习步长:

$$\lambda_1 = \frac{W_{ik}^{(n)}}{(WHH^T)_{ik}}, \lambda_2 = \frac{H_{kj}^{(n)}}{(W^TW H)_{kj}}$$

将学习步长代入得:

$$W_{ik} \leftarrow W_{ik} \frac{(VH^T)_{ik}}{(WHH^T)_{ik}} \qquad H_{kj} \leftarrow H_{kj} \frac{(W^TV)_{kj}}{(W^TW H)_{kj}}$$

Algorithm NMF_Euclid

INPUT: V

OUTPUT: W, H

Randomize W, H ;

Repeat

Update $W_{ik} \leftarrow W_{ik} \frac{(V H^T)_{ik}}{(W H H^T)_{ik}}$

Update $H_{kj} \leftarrow H_{kj} \frac{(W^T V)_{kj}}{(W^T W H)_{kj}}$

Until converge or maximal iterations

3.2 第二种损失函数：基于KL散度的损失函数

$$\min_{W,H} \sum_{i,j} (V_{ij} \log \frac{V_{ij}}{(WH)_{ij}} - V_{ij} + (WH)_{ij})$$

Kullback–Leibler Divergence

$$D_{\text{KL}}(P\|Q) = \sum_i P(i) \ln \frac{P(i)}{Q(i)}$$

Information Entropy

$$H(X) = \sum_i P(x_i) I(x_i) = - \sum_i P(x_i) \log_b P(x_i),$$

3 Cost functions

To find an approximate factorization $V \approx WH$, we first need to define cost functions that quantify the quality of the approximation. Such a cost function can be constructed using some measure of distance between two non-negative matrices A and B . One useful measure is simply the square of the Euclidean distance between A and B [13],

$$\|A - B\|^2 = \sum_{ij} (A_{ij} - B_{ij})^2 \quad (2)$$

This is lower bounded by zero, and clearly vanishes if and only if $A = B$.

Another useful measure is

$$D(A||B) = \sum_{ij} \left(A_{ij} \log \frac{A_{ij}}{B_{ij}} - A_{ij} + B_{ij} \right) \quad (3)$$

Like the Euclidean distance this is also lower bounded by zero, and vanishes if and only if $A = B$. But it cannot be called a “distance”, because it is not symmetric in A and B , so we will refer to it as the “divergence” of A from B . **It reduces to the Kullback-Leibler divergence, or relative entropy, when $\sum_{ij} A_{ij} = \sum_{ij} B_{ij} = 1$, so that A and B can be regarded as normalized probability distributions.**

- 假设噪声E服从泊松分布：

$$p(V_{ij}|W, H) = \frac{(WH)_{ij}^{X_{ij}}}{X_{ij}!} \exp(-(WH)_{ij})$$

- 构造似然函数：

$$L(W, H) = \sum_{ij} [V_{ij} \log(WH)_{ij} - (WH)_{ij} - \log(X_{ij}!)]$$

- 更新规则:

$$W_{ik} \leftarrow W_{ik} \frac{\sum_j H_{kj} X_{ij} / (WH)_{ij}}{\sum_j H_{kj}}$$

$$H_{kj} \leftarrow H_{kj} \frac{\sum_i W_{ik} X_{ij} / (WH)_{ij}}{\sum_i W_{ik}}$$

3.3 扩展应用1：疾病-基因组双向聚类

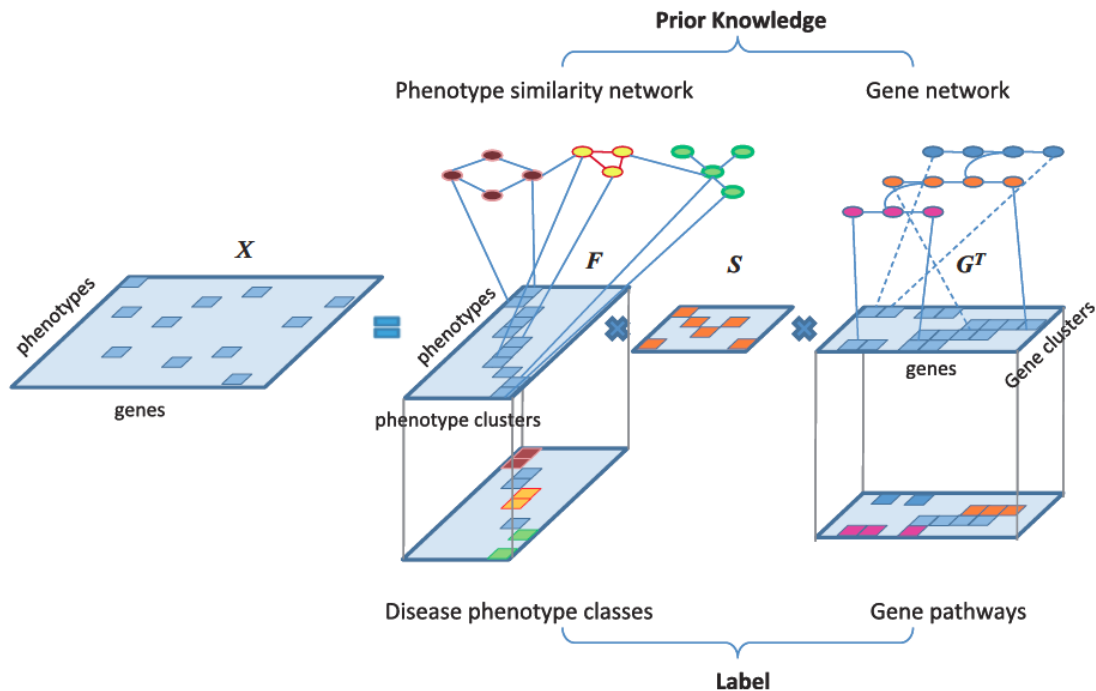


Figure 1. NMTF of disease phenotype-gene associations. The phenotype-gene association matrix X is factorized into products of three matrices, phenotype cluster membership F , gene cluster membership G and phenotype cluster-gene cluster association S for supervised co-clustering of phenotypes and genes. Label information for the disease classes and the pathways are available for a small number of phenotypes and genes. Prior knowledge is also introduced from phenotype similarity network and gene network. For better visualization, different colors are used to distinguish the phenotypes and the genes in different clusters.

加入已知约束的代价函数

$$\min_{F,S,G} \|X - FSG^T\|_F^2 + \alpha \|F - F^0\|_F^2 + \beta \|G - G^0\|_F^2$$

$$\text{subject to } \sum_{j=1}^{k_1} F_{i,j} = 1, \sum_{j=1}^{k_2} G_{i,j} = 1.$$

- 矩阵分解误差
- 疾病聚类同先验吻合
- 基因聚类同先验吻合

Table 1. Notations

Notation	Definition
m	Number of disease phenotypes
n	Number of genes
k_1	Number of phenotype clusters (e.g. classes)
k_2	Number of gene clusters (e.g. pathways)
X	Disease phenotype–gene association matrix ($m \times n$)
F	Phenotype cluster membership ($m \times k_1$)
S	Phenotype cluster–gene cluster association Matrix ($k_1 \times k_2$)
G	Gene cluster membership ($n \times k_2$)
F^0	Annotated phenotype cluster membership ($m \times k_1$)
G^0	Annotated gene cluster membership ($n \times k_2$)
M	Disease phenotype similarity network ($m \times m$)
N	Gene interaction network ($n \times n$)

Regularized Non-negative Matrix Tri-factorization

INPUT: X , F^0 , G^0 , L_M , L_N , parameters α , β , γ , and λ , maximum iteration T

OUTPUT: F , G , S

while not converged and $t \leq T$ **do**

(1) Update $F_{ij} \leftarrow F_{ij} \sqrt{\frac{(XGS^T + \alpha F^0 + \gamma MF)_{ij}}{(FSG^T GS^T + \alpha F + \gamma D_M F)_{ij}}}.$

(2) Normalize $F_{i.} \leftarrow \frac{F_{i.}}{\sum_{j=1}^{k_1} F_{ij}}.$

(3) Update $G_{ij} \leftarrow G_{ij} \sqrt{\frac{(X^T FS + \beta G^0 + \lambda NG)_{ij}}{(GS^T F^T FS^S + \beta G + \lambda D_N G)_{ij}}}.$

(4) Normalize $G_{.j} \leftarrow \frac{G_{.j}}{\sum_{i=1}^{k_2} G_{ij}}.$

(5) Compute $S_{ij} \leftarrow S_{ij} \sqrt{\frac{(F^T XG)_{ij}}{(F^T FSG^T G)_{ij}}}.$

end while

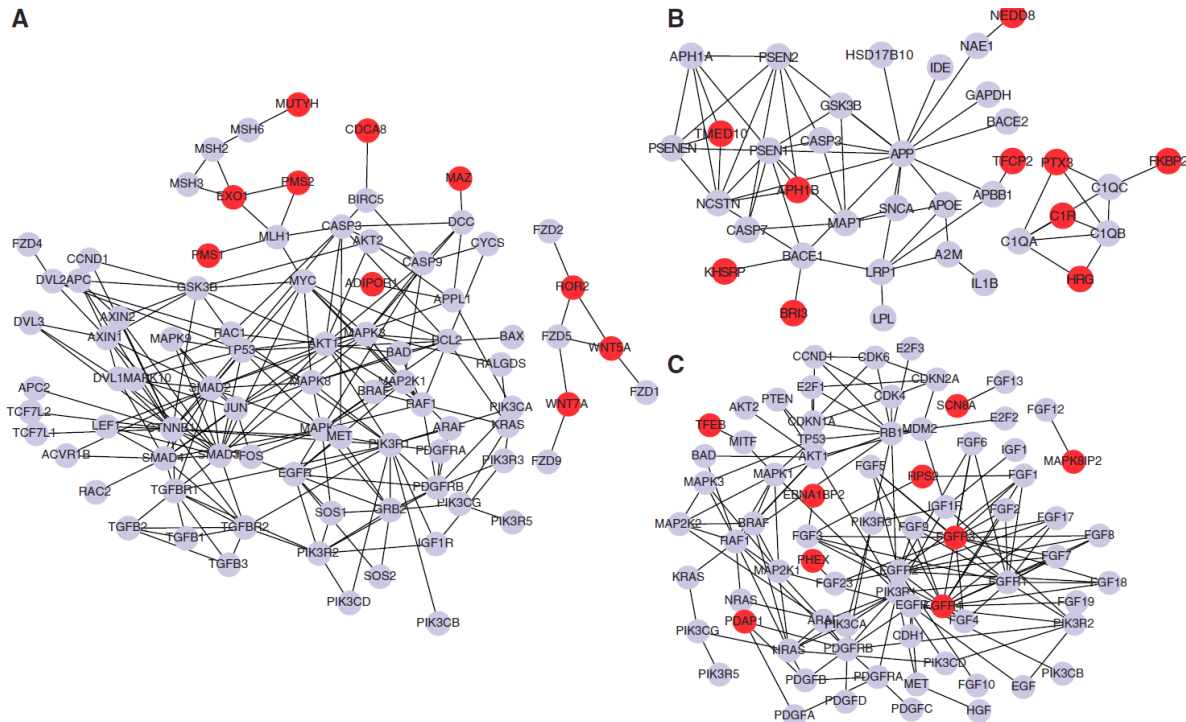


Figure 5. PPI subnetworks of the extended disease pathways. In each pathway, gray nodes are known member genes in the disease pathways and red nodes are newly predicted member genes. Edges represent PPI between two genes. Note that if a known or a newly predicted member gene is not interacting with any other member genes in the pathway, the gene is not included. **(A)** Colorectal cancer pathway. The predicted colorectal cancer genes EXO1 and ADIPOR1 are interacting with many other genes in the colorectal cancer pathway. **(B)** Alzheimer pathway. Over-expression of C1R is known for involving alzheimer disease. **(C)** Melanoma pathway. Mutation and copy number changes in new member gene FGFR3 were recently discovered in melanoma.

3.4 扩展应用2：带有层次约束的多矩阵分解

一致性矩阵分解

不同粒度描述表征不影响聚类结果

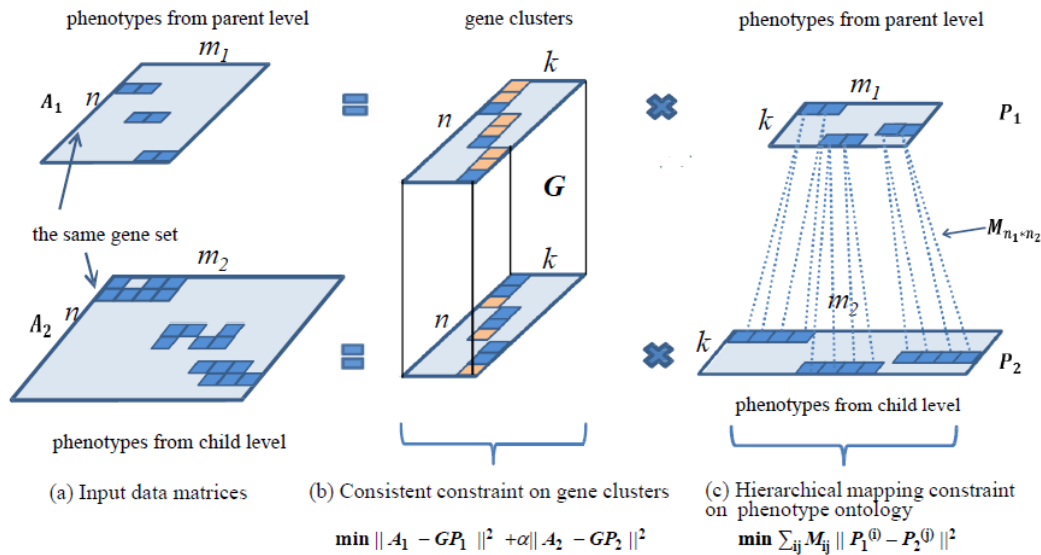
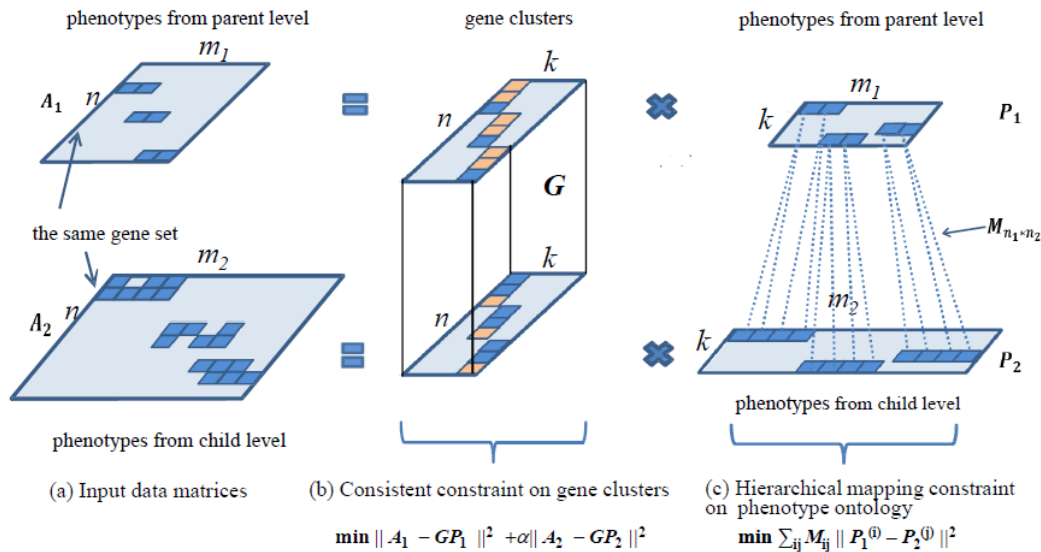


Figure 1: Illustration of the CMNMF framework. (a) The gene-phenotype associations are divided into two matrices according to the level of phenotype ontologies, and the two matrices share the same gene set. (b) consistent constraint on factorized gene clusters. (c) Hierarchical mapping constraint on the phenotype ontologies at parent and child levels.

帶有一致性约束的代价函数



$$L = \|A_1 - GP_1\|_F^2 + \alpha \|A_2 - GP_2\|_F^2 + \beta \sum_{ij} M_{ij} \|P_1^{(i)} - P_2^{(j)}\|^2$$

$$\text{s.t. } G \geq 0, P_1 \geq 0, P_2 \geq 0$$

Algorithm 1 CMNMF

Input: A_1 : gene-phenotype association matrix at parent level

A_2 : gene-phenotype association matrix at child level

α, β : hyper-parameters

Output: model parameters G, P_1, P_2

1: $G, P_1, P_2 \leftarrow$ random values

2: **repeat**

3: Update $G_{ij} \leftarrow G_{ij} \frac{(A_1 P_1^T + \alpha A_2 P_2^T)_{ij}}{(G P_1 P_1^T + \alpha G P_2 P_2^T)_{ij}}$

4: Update $(P_1)_{ij} \leftarrow (P_1)_{ij} \frac{(G^T A_1 + \beta P_2 M^T)_{ij}}{(G^T G P_1 + \beta P_1 D_1)_{ij}},$

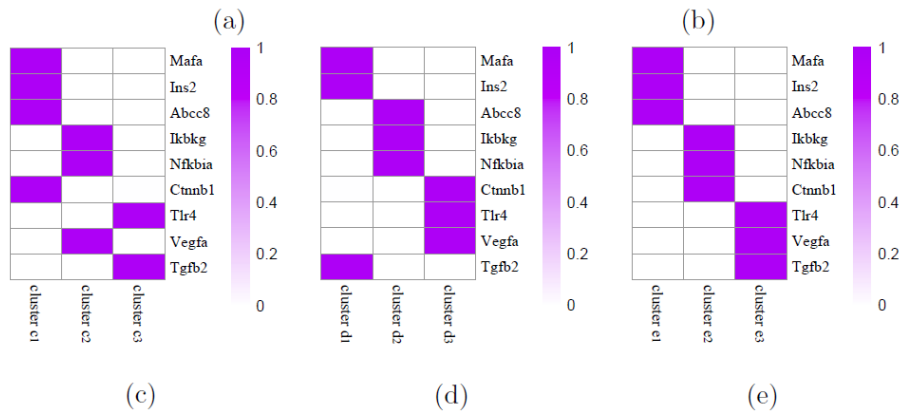
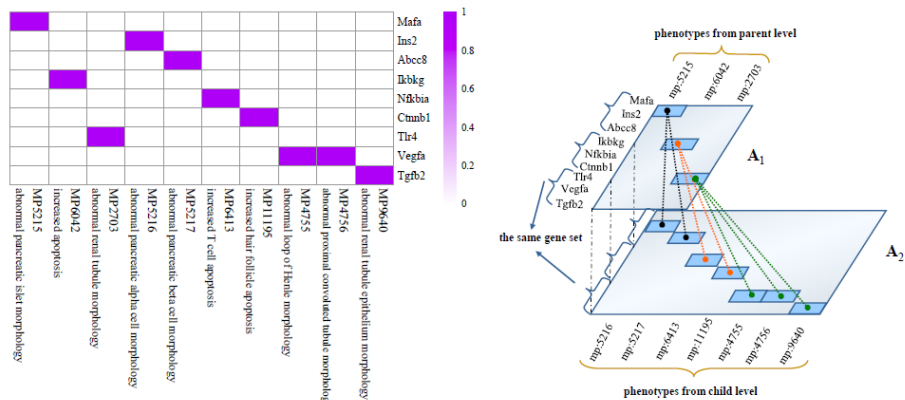
 Update $(P_2)_{ij} \leftarrow (P_2)_{ij} \frac{(\alpha G^T A_2 + \beta P_1 M)_{ij}}{(\alpha G^T G P_2 + \beta P_2 D_2)_{ij}}$

5: **until** convergence

6: Normalize $G_{ik} \leftarrow \frac{G_{ik}}{\sqrt{\sum_i G_{ik}^2}}$

7: Normalize $(P_1)_{kj} \leftarrow (P_1)_{kj} \sqrt{\sum_i G_{ik}^2}, \quad (P_2)_{kj} \leftarrow (P_2)_{kj} \sqrt{\sum_i G_{ik}^2}$

8: **return** G, P_1, P_2



1. 距离、空间、KNN分类器
2. 主成分分析(Principle Component Analysis)
3. 非负矩阵分解(Non-negative Matrix Factorization)